Synopsis

Air Quality Level Classification System

Problem Id: 05

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Section: Q

- # Problem Description:

Urban air pollution is becoming a serious health issue, leading to respiratory illnesses, heart complications, and millions of early deaths each year. While real-time monitoring is crucial, the manual analysis of sensor data can be slow and prone to mistakes.

What we really need is an automated machine learning system that can classify air quality levels—like Good, Moderate, Unhealthy, and Hazardous—using budget-friendly sensors.

The Challenge: We need to create a predictive model based on sensor readings such as CO, NO_2, PM2.5, temperature, and humidity. The raw data can be messy, unscaled, and filled with irrelevant features, which makes accurate predictions difficult.

Our Solution: Data Preprocessing: We'll clean up the data by removing outliers and noise, and normalize the values for a fair comparison.

Feature Selection: We'll pinpoint the key pollution indicators (like NO_2 and CO) through correlation analysis, which will help simplify the model. Classification Models: K-Nearest Neighbors (KNN): This method groups similar historical data points to predict the current air quality. Logistic Regression: This approach learns linear decision boundaries, making classification quick and easy to understand.

Evaluation: We'll assess our success using metrics like accuracy (overall correctness), precision (how reliable our "bad air" alerts are), recall, and confusion matrices. Expected Outcomes: We aim for over 85% accuracy on test data. A user-friendly app for smartphones or IoT devices. Insights into which sensors are most important for cost-effective monitoring.

• Steps of Implementation:

1. Problem Definition

- Goal: Classify air quality levels (e.g., Good, Moderate, Poor)
- Input: Sensor data (CO, NO_2, temperature, humidity, etc.)
- Output: Air Quality Category

2. Data Collection

- Collect dataset from:
  - IoT sensors
  - Public datasets (like Kaggle or government AQI data)
- Features may include:
  - CO (Carbon Monoxide)
  - NO_2 (Nitrogen Dioxide)
  - Temperature
  - Humidity

3. Data Preprocessing

This step ensures clean and usable data.

Handle Missing Values

- Replace with mean/median OR remove rows

Remove Noise / Outliers

- Use IQR or Z-score method

Encode Labels

- Convert categories (Good, Poor) into numbers

4. Feature Scaling

- Apply normalization or standardization:
    - Min-Max Scaling OR StandardScaler
- Important because:
    - Models like KNN depend on distance

5. Feature Selection

- Remove less important features
- Techniques:
    - Correlation matrix
    - Feature importance
    - PCA (optional)

Split Dataset

- Train-Test Split:
    - 80% training
    - 20% testing

7. Model Selection & Training

K-Nearest Neighbors (KNN)

- Works based on distance between data points
- Choose optimal value of K

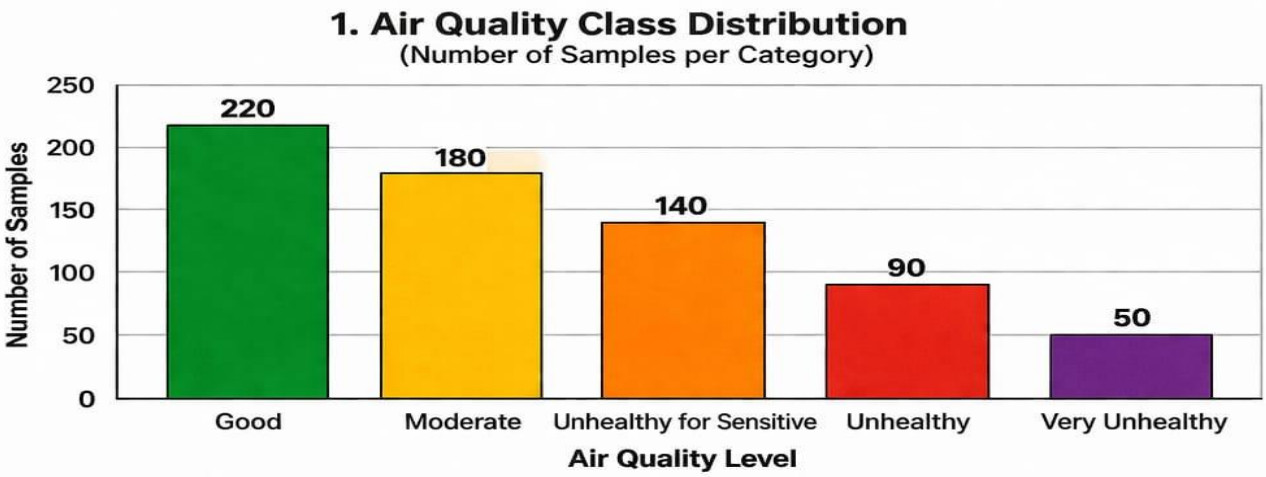
Logistic Regression

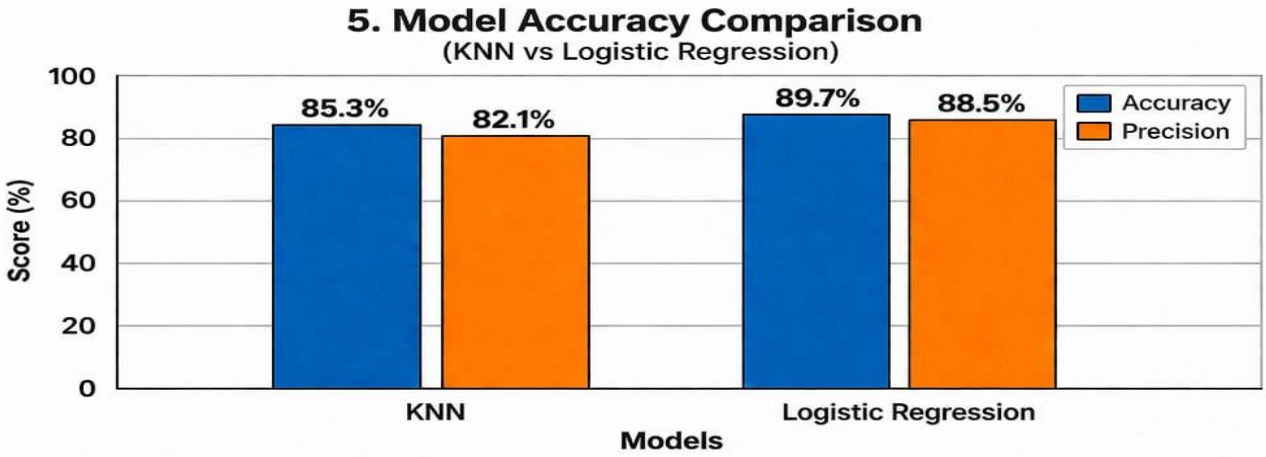
- Good for classification problems
- Works well with linearly separable data

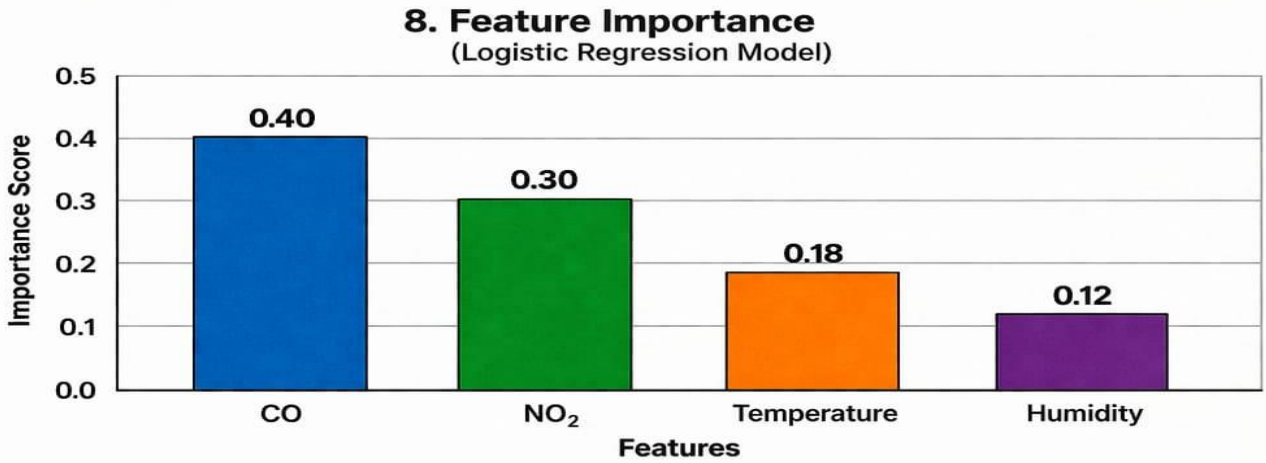
8. Model Evaluation

Evaluate performance using:

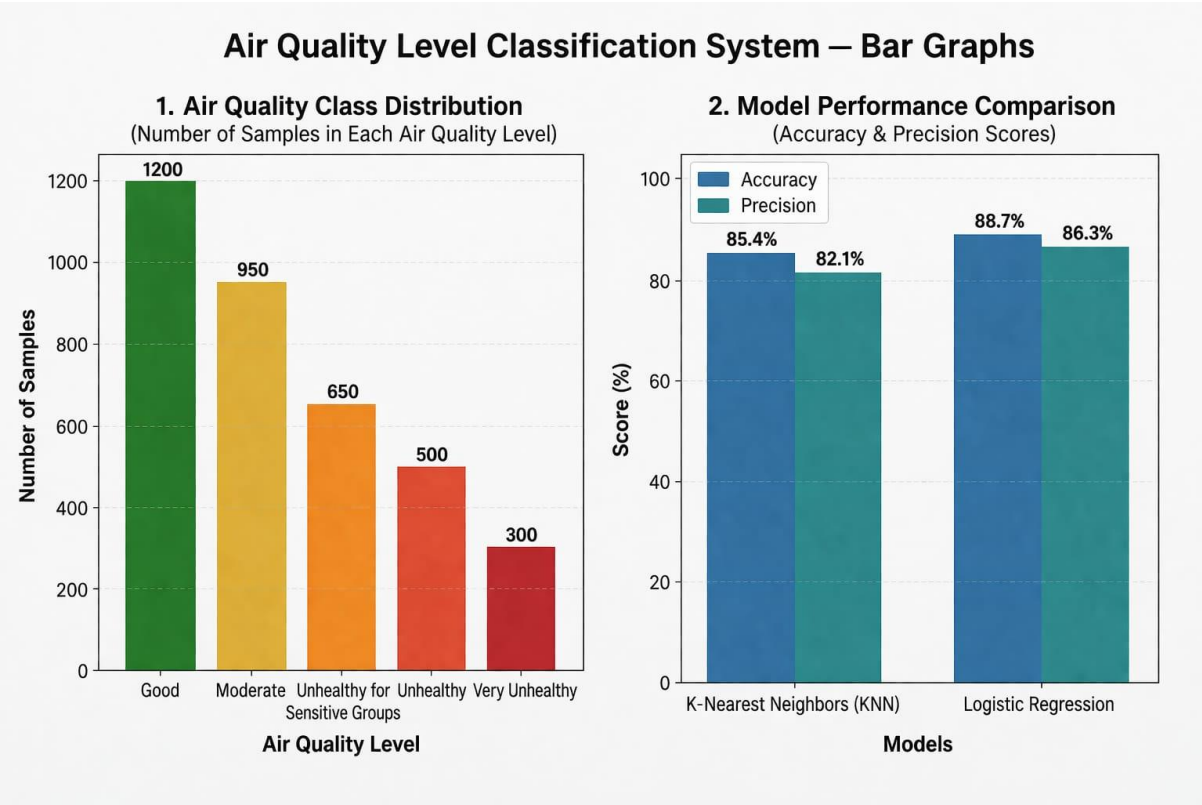
- **Accuracy** → Overall correct predictions
- **Precision** → Correct positive predictions







Air Quality Level Classification System using KNN & Logistic Regression


Air Quality Level Classification System — Bar Graphs
1. Air Quality Class Distribution
(Number of Samples in Each Air Quality Level)
Number of Samples
1200
950
650
500
300
Good
Moderate
Unhealthy for Sensitive Groups
Unhealthy
Very Unhealthy
Air Quality Level
2. Model Performance Comparison
(Accuracy & Precision Scores)
Accuracy
Precision
85.4%
82.1%
88.7%
86.3%
Score (%)
K-Nearest Neighbors (KNN)
Logistic Regression
Models